

Histology of skin & eyelid



MANIPAL
UNIVERSITY COLLEGE
MALAYSIA

Formerly known as  **MELAKA-MANIPAL**
MEDICAL COLLEGE
MALAYSIAN ACADEMY OF HIGHER EDUCATION

Dr. Sudhakara Babu Ch.
Associate Professor, Department of Anatomy,
MUCM



#inspire

Introduction

- The skin is the heaviest organ in the body (about 16% of body weight).
- The skin consists of :
 1. Epidermis
 2. Dermis

Thin skin



Thick skin



Types of Skin

- There are two types of skin depending on thickness of epidermis:

❑ Thick skin:

- Thick skin has very thick layer of epidermis – shows thick stratum corneum – also called glabrous skin
- Does not have hairs.

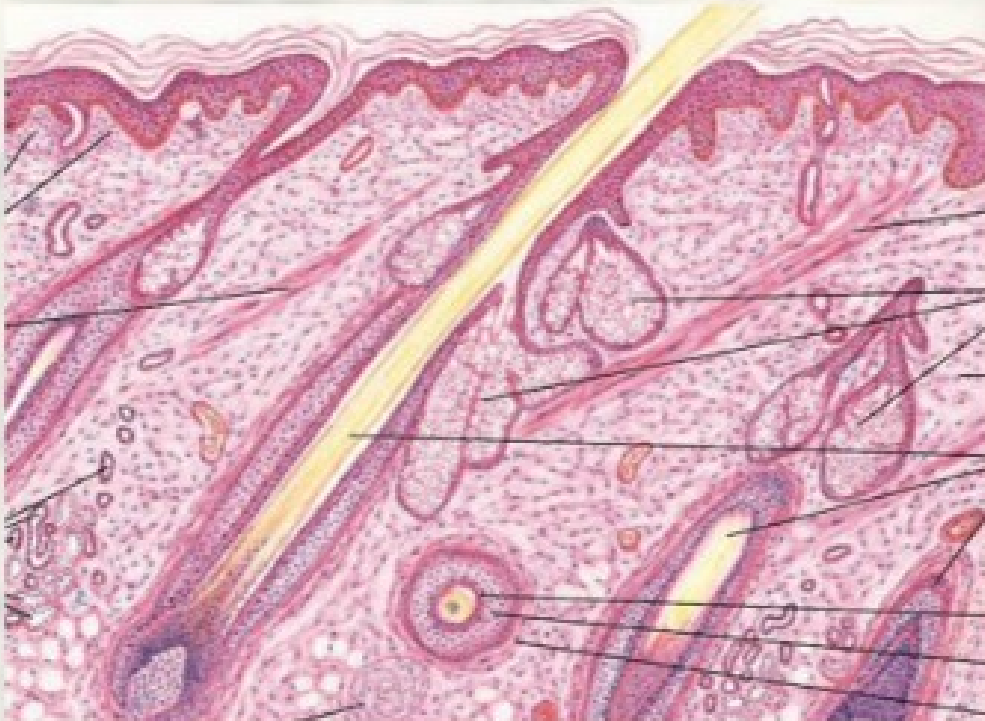
Locations: Skin of palm of hand and sole of feet

❑ Thin (hairy) skin: Epidermis is thin – shows hairs.

Locations: Skin covering all parts of body except palm and soles

Types of Skin

1- Thin Skin



2- Thick Skin



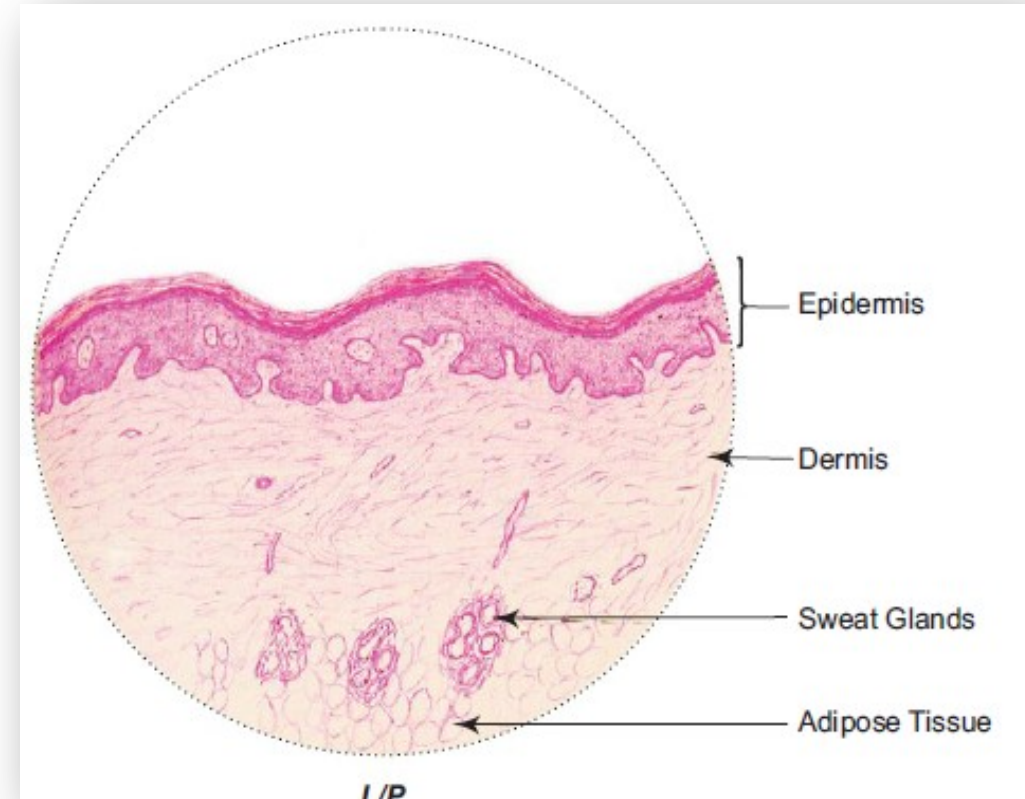
Structure of Skin

❑ Histologically, skin consists of two layers:

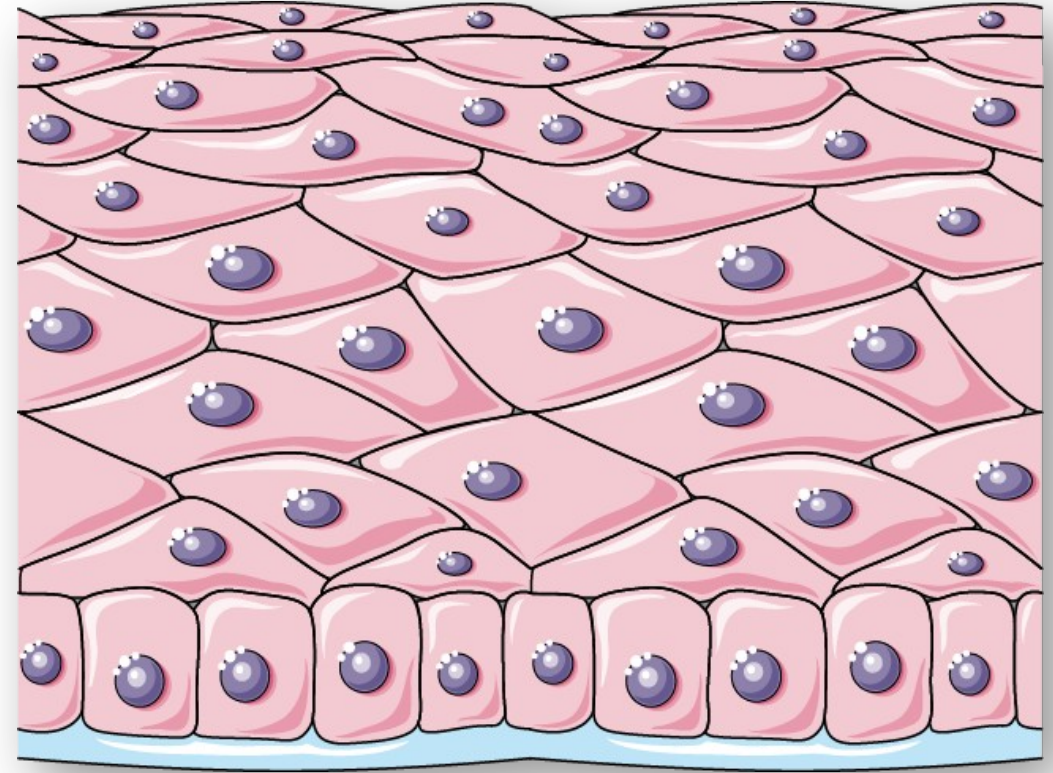
- 1. Epidermis:** Superficial layer that consists of keratinized stratified squamous epithelium
- 2. Dermis:** Deeper layer that is made up of connective tissue

1. Epidermis

- Stratified squamous keratinized epithelium
- Projects into dermis as epidermis ridges
- Avascular
- Nerve endings are seen its basal layer
- Renewed for every 15-30 days



- Consists of keratinized stratified squamous epithelium
- Shows five layers of cells as follows :
 1. Stratum basale
 2. Stratum spinosum
 3. Stratum granulosum
 4. Stratum lucidum (only in thick skin)
 5. Stratum corneum



Layers of epidermis

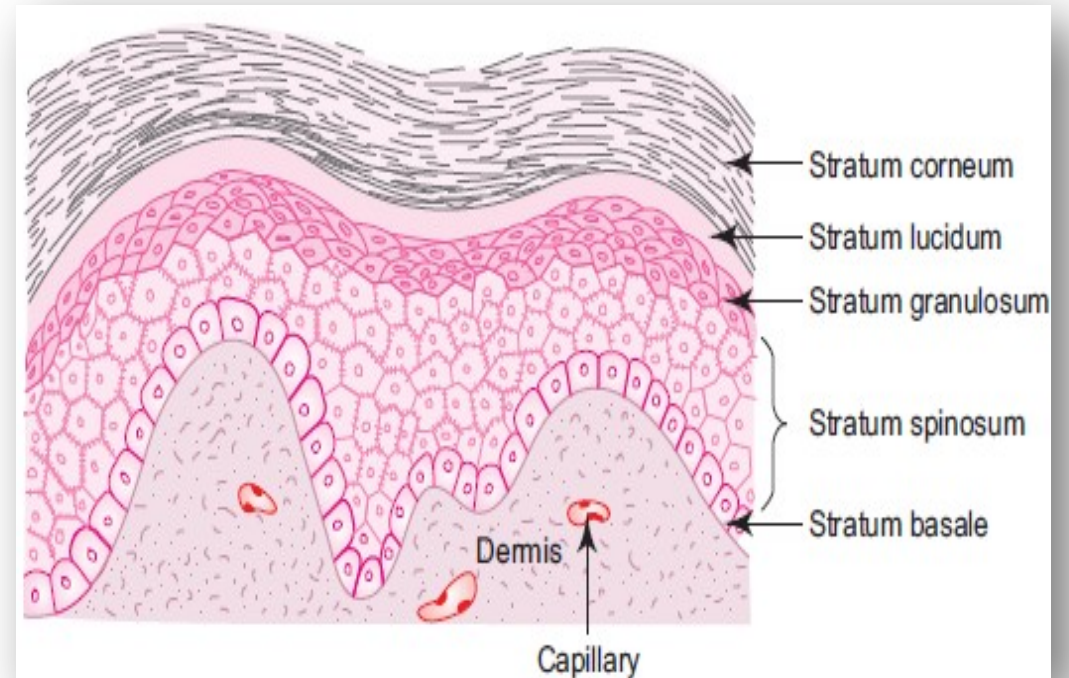
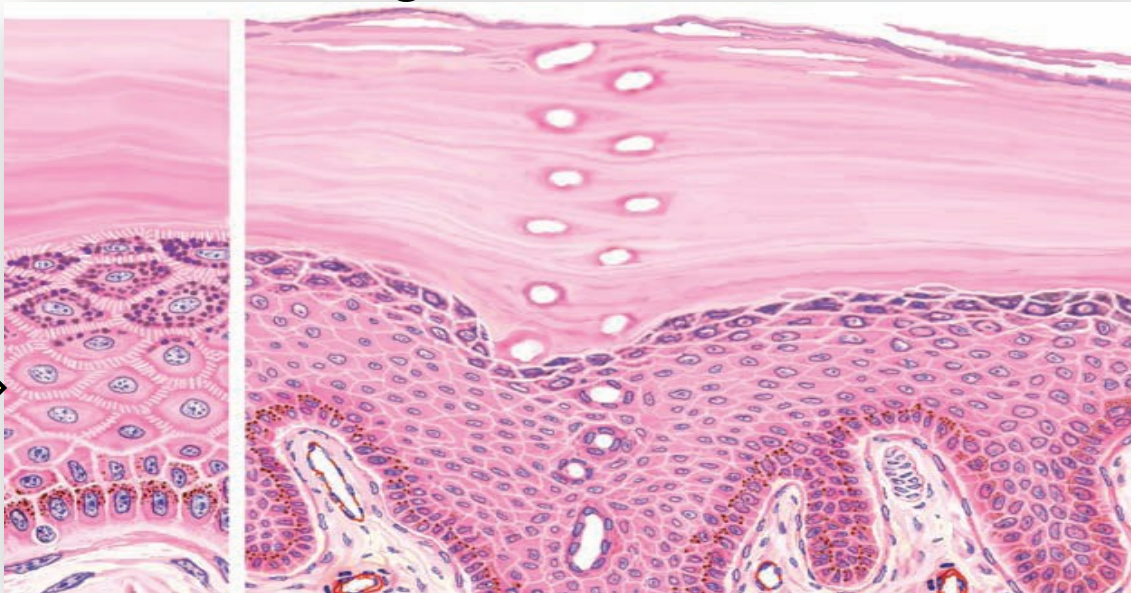
□ Stratum basale (Stratum Germinativum)

- Deepest layer of epidermis
- *Single layer of cuboidal (low columnar) cells* that rest on basal lamina & contains melanocytes.
- Cells divide mitotically to give rise to keratinocytes (cells of skin) that form superficial layers of skin
- Hence, called stratum germinativum.



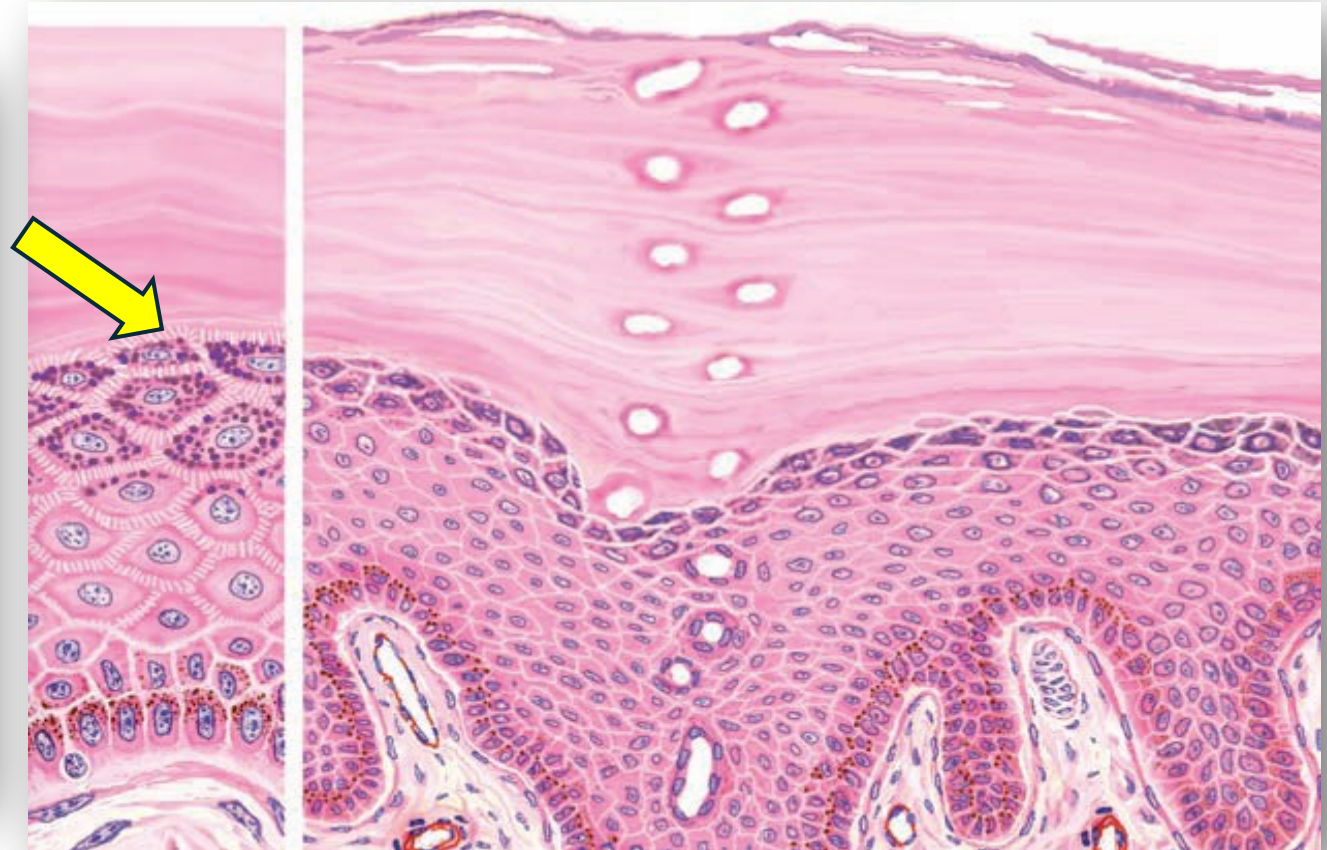
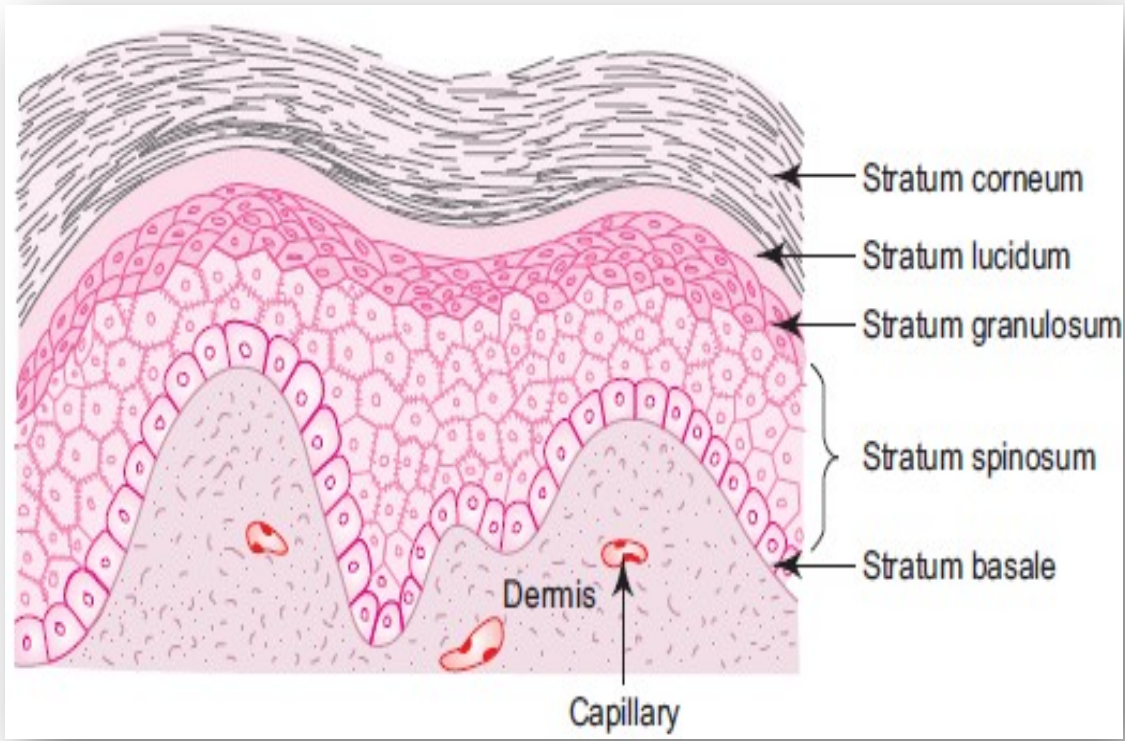
❑ Stratum spinosum (Malpighian layer)

- Polyhedral cells connected by desmosomes.
- Keratinocytes in this layer shows small spine-like projections on their external surface. (due to artefact while preparing the slide for sectioning)
- Some cells of may undergo mitosis; hence, stratum spinosum & stratum basale together can be considered as germinative zone.



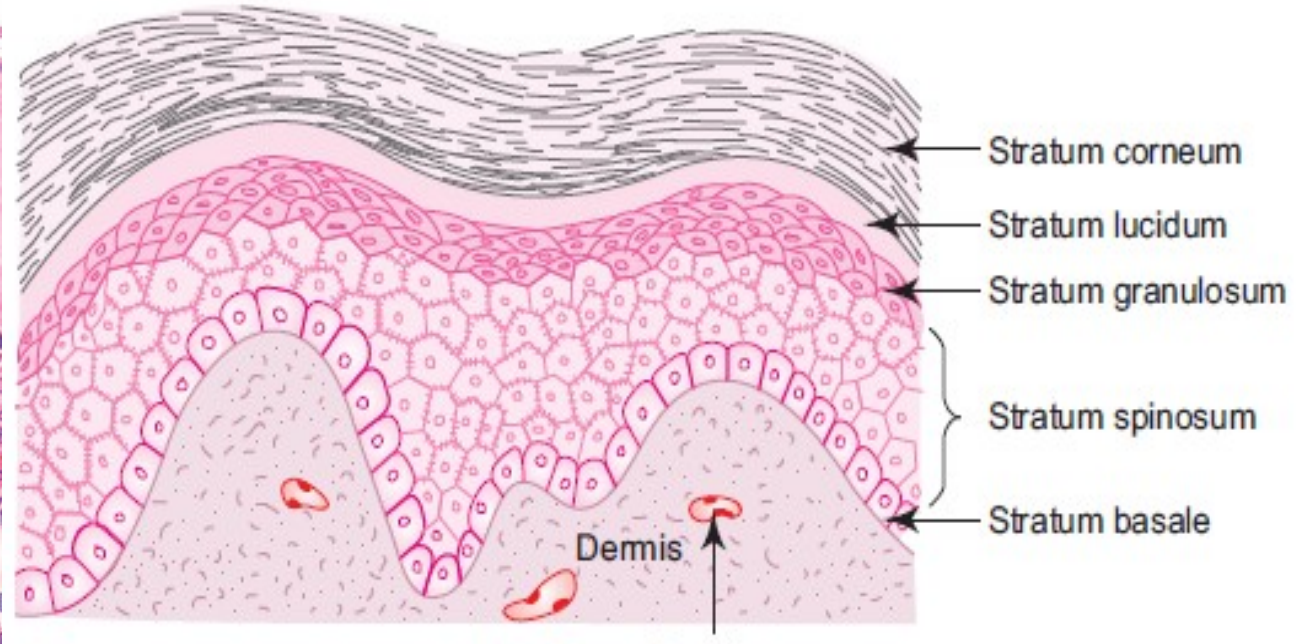
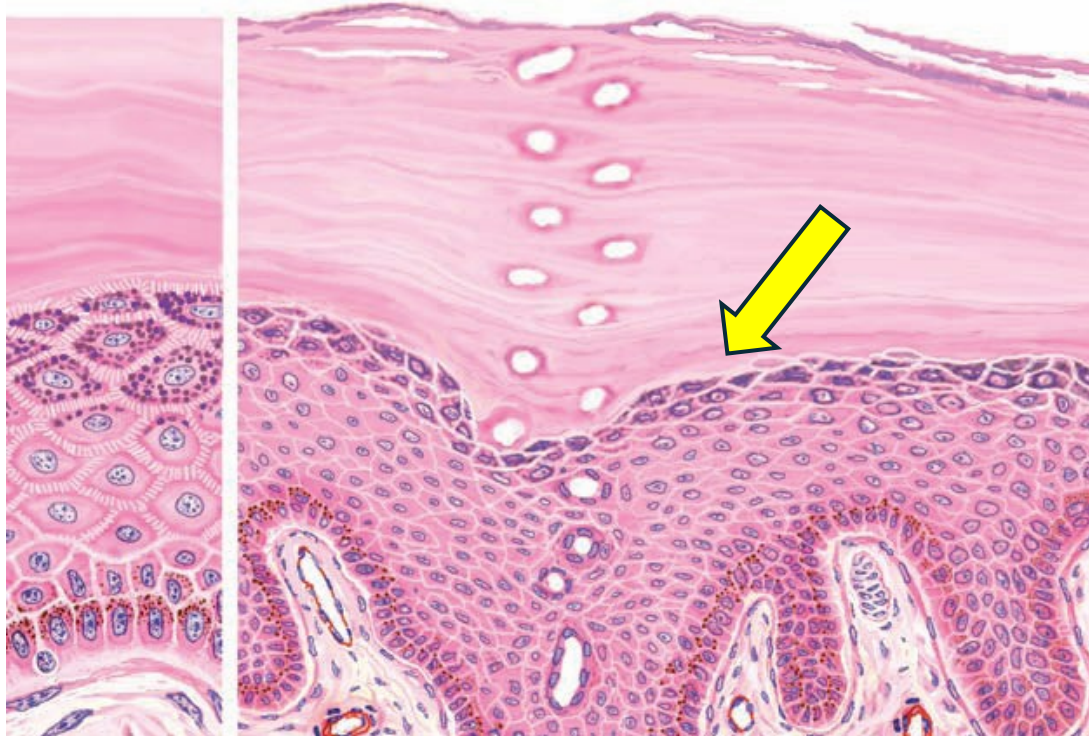
❑ Stratum granulosum

- 3-5 layers of flattened polyhedral cells
- Cells are filled with basophilic keratohyalin granules.



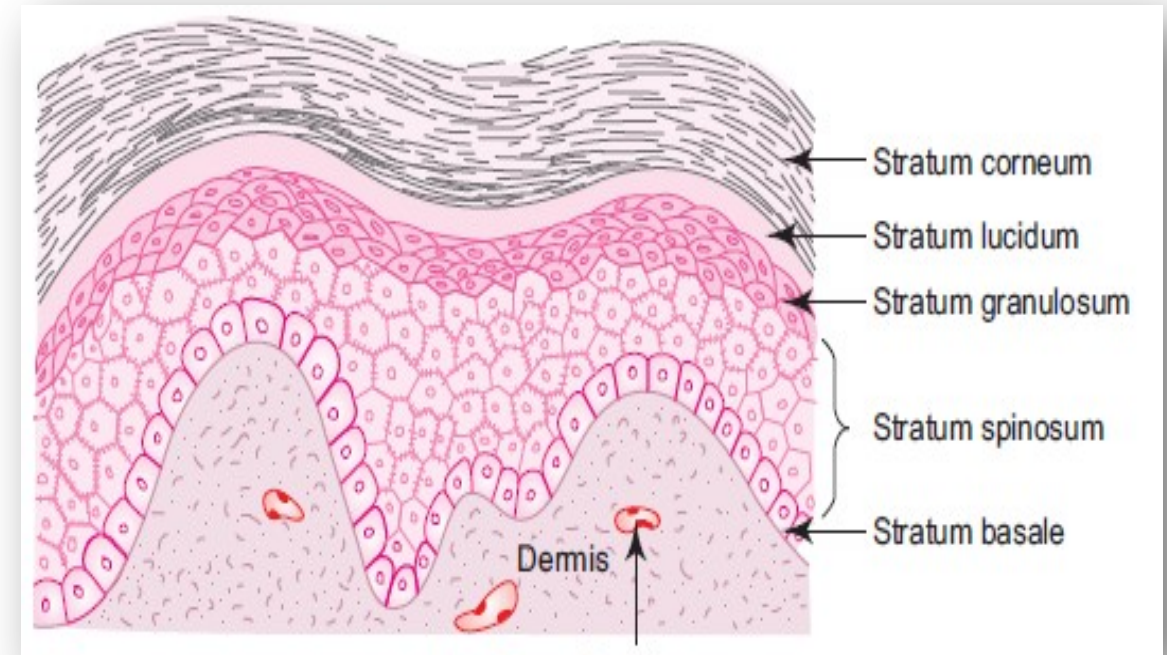
❑ Stratum lucidum

- Seen only in thick skin.
- Stained lightly with eosin and appears homogenous.
- Flattened eosinophilic dead cells
- No organelles and nuclei
- Cytoplasm is filled with tough scleroprotein - keratin



❑ Stratum corneum

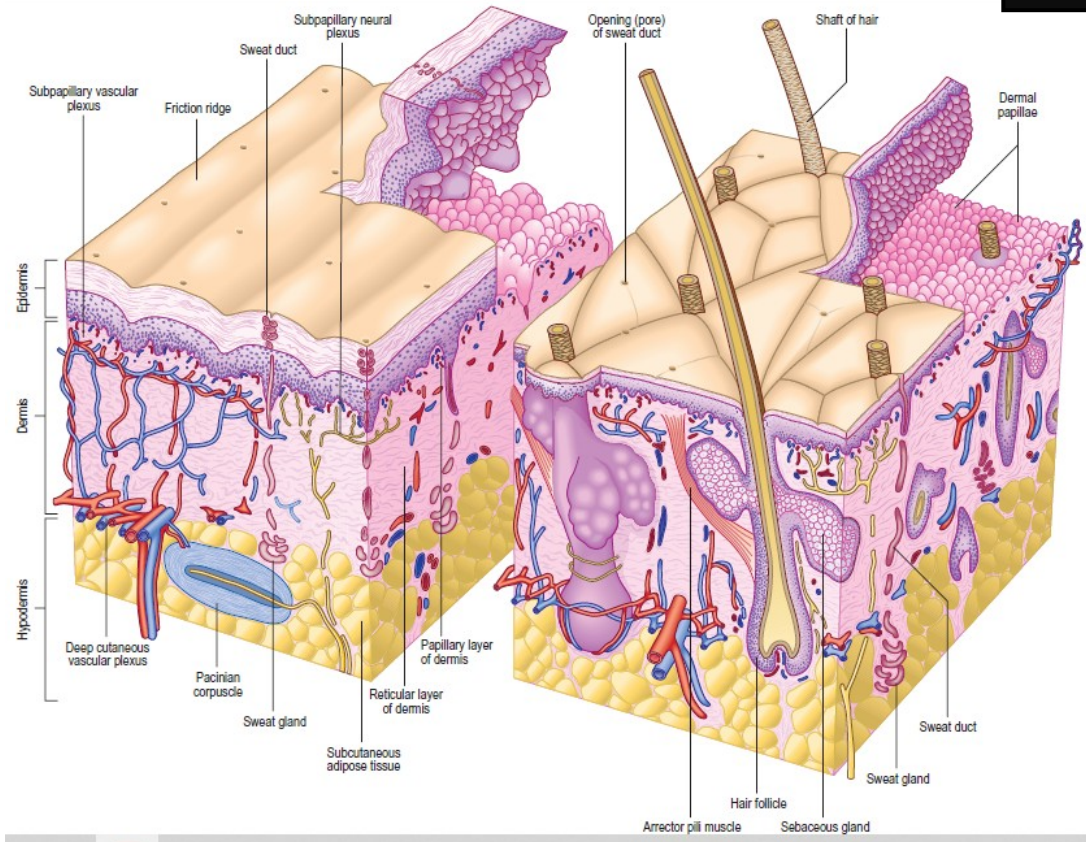
- Most superficial layer of epidermis.
- Flattened non-nucleated, scaly like dead keratinized cells without any organelles.
- Cytoplasm is filled with keratin.
- Superficial cells are being constantly shed off and are replaced by proliferation of stratum germinativum



CELLS OF EPIDERMIS

- Skin has four types of cells:
 - Keratinocyte
 - Melanocytes,
 - Langerhans cells
 - Merkel's cells

2.Dermi



- **S** Made up of vascular connective tissue
- Corresponds to lamina propria of mucous membrane
- It is divided into
 - Papillary layer
 - Reticular layer

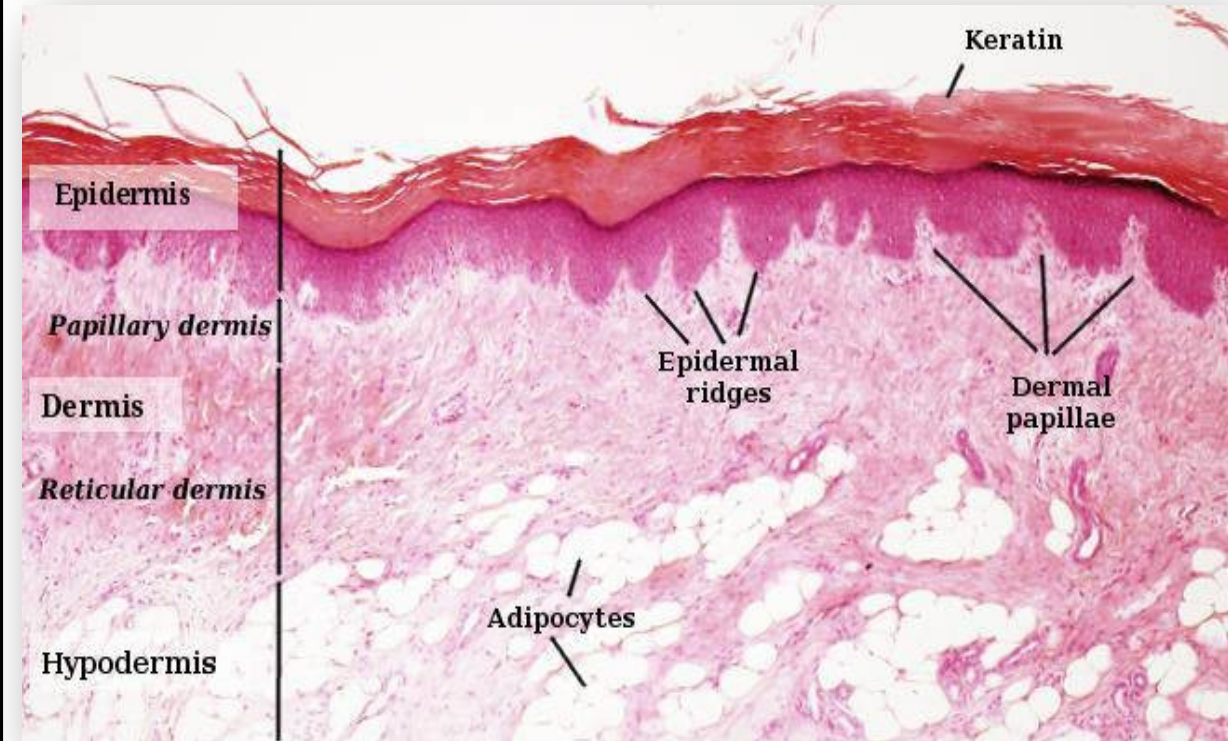
Layers of the dermis

❖ Papillary layer:

- Narrow band of loose connective tissue.
- In contact with basement membrane of stratum basale.
- Dermal papillae (finger- like processes)

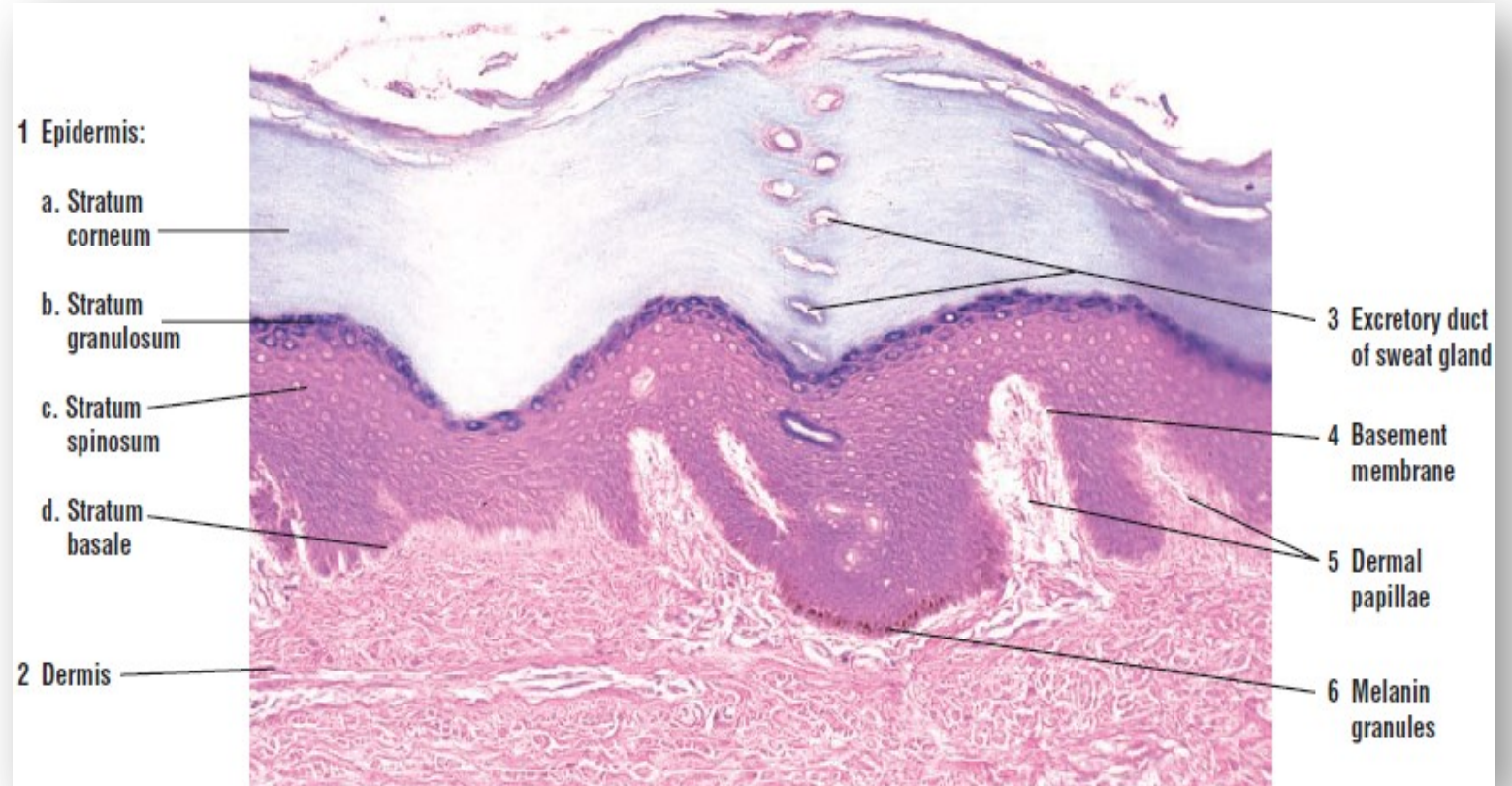
❖ Reticular layer:

- Dense irregular connective tissue.
- Thick collagen fibres.



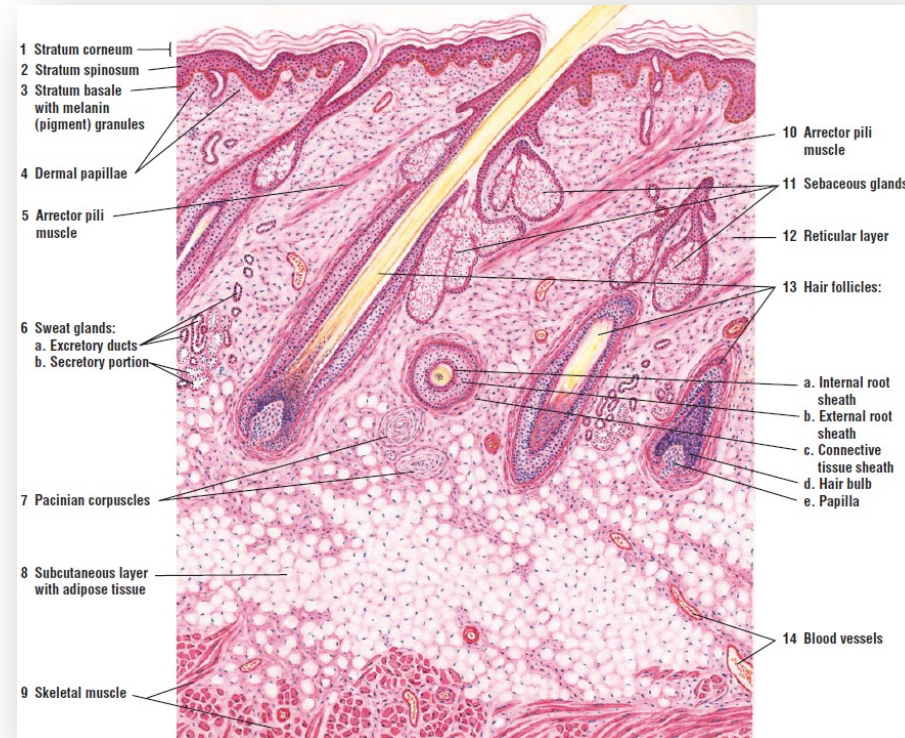
Thick skin

- Stratum corneum is very thick.
- Dermal papillae more prominent to resist shearing frictional forces.
- Absence of hair follicles, sebaceous glands & arrector pilorum muscle.
- Presence of sweat glands.



Thin skin

- Thin layer of stratum corneum
- Thickness of other layers is reduced
- Dermal papillae less prominent due to reduced friction.
- Stratum Lucidum is absent
- Presence of hair follicles, sebaceous glands, sweat glands, arrector pilorum muscle.



Differences between thin & thick skin

	Thick skin	Thin skin
Stratum lucidum	Present	Absent
Thickness of epidermis	0.07–1.5 mm	0.01–0.15 mm
Epidermal ridges	Present	Absent
Hair follicles	Absent	Present
Arrector pili muscle	Absent	Present
Sebaceous glands	Absent	Present
Sweat glands	Many	Few
Sensory receptors/Merkel's cells	More	Less
Location	Skin of palm and sole	Skin except of palm and sole

Eyelid

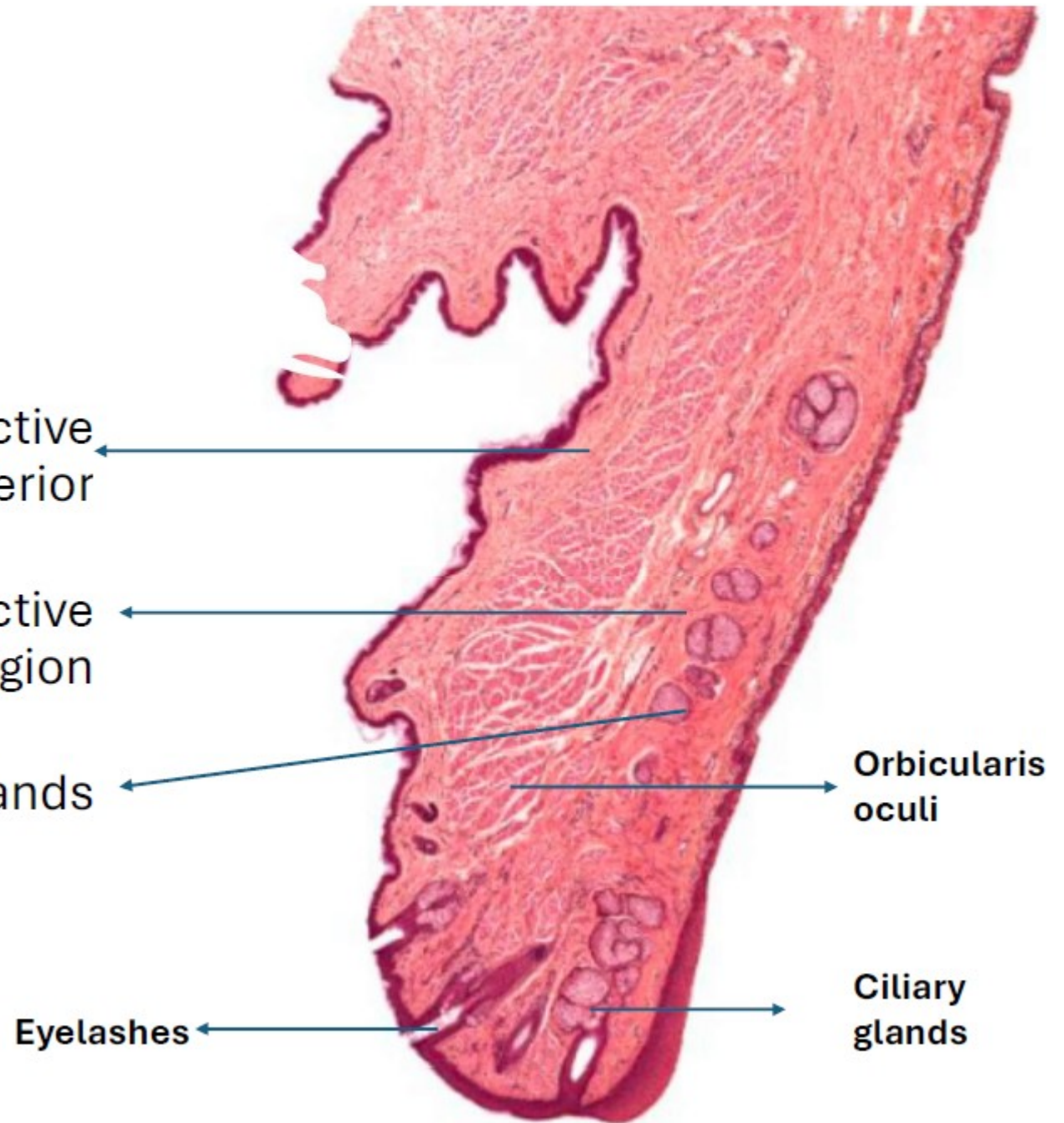
- Each eyelid (or tarsus) consists of a thin envelope of tissue that protects the eye by covering the cornea.
- **Anterior surface:** covered by thin skin (stratified squamous keratinized only a few cell layers thick).
- **Posterior surface:** covered by **conjunctiva** which is a stratified columnar epithelium with numerous mucus-producing goblet cells.



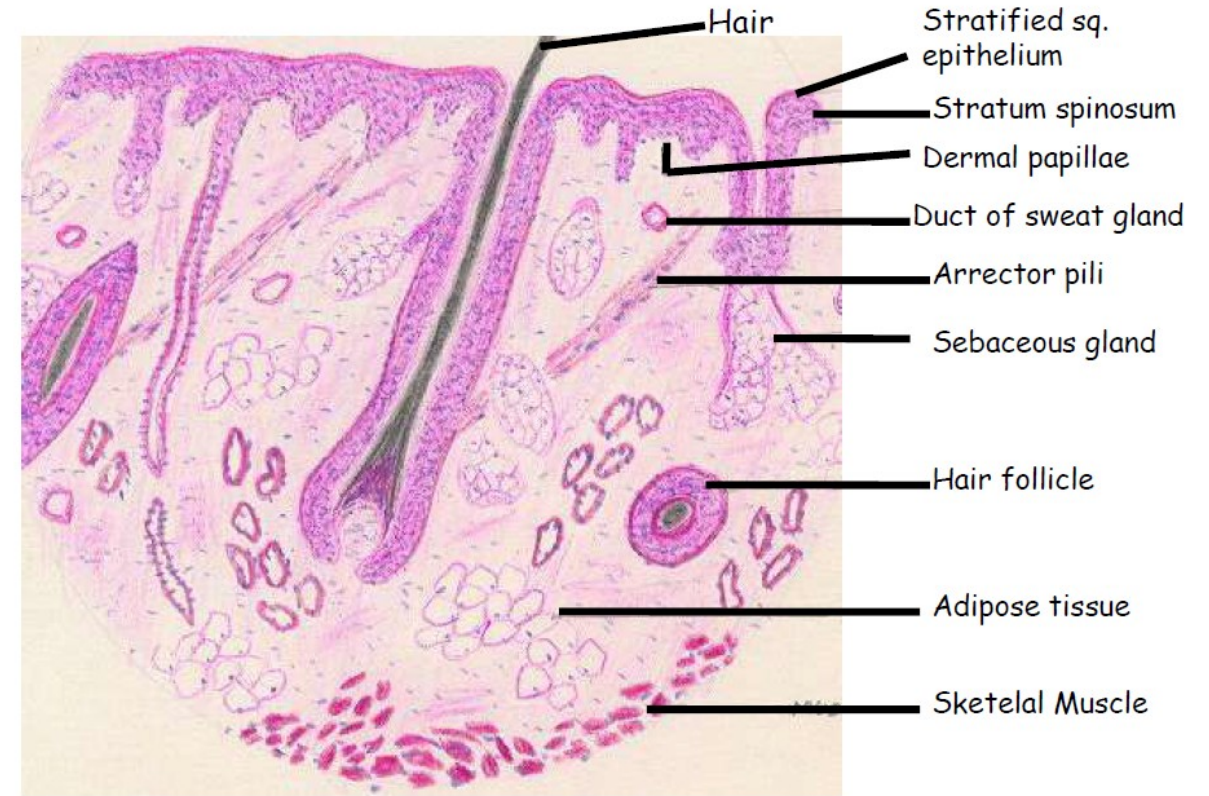
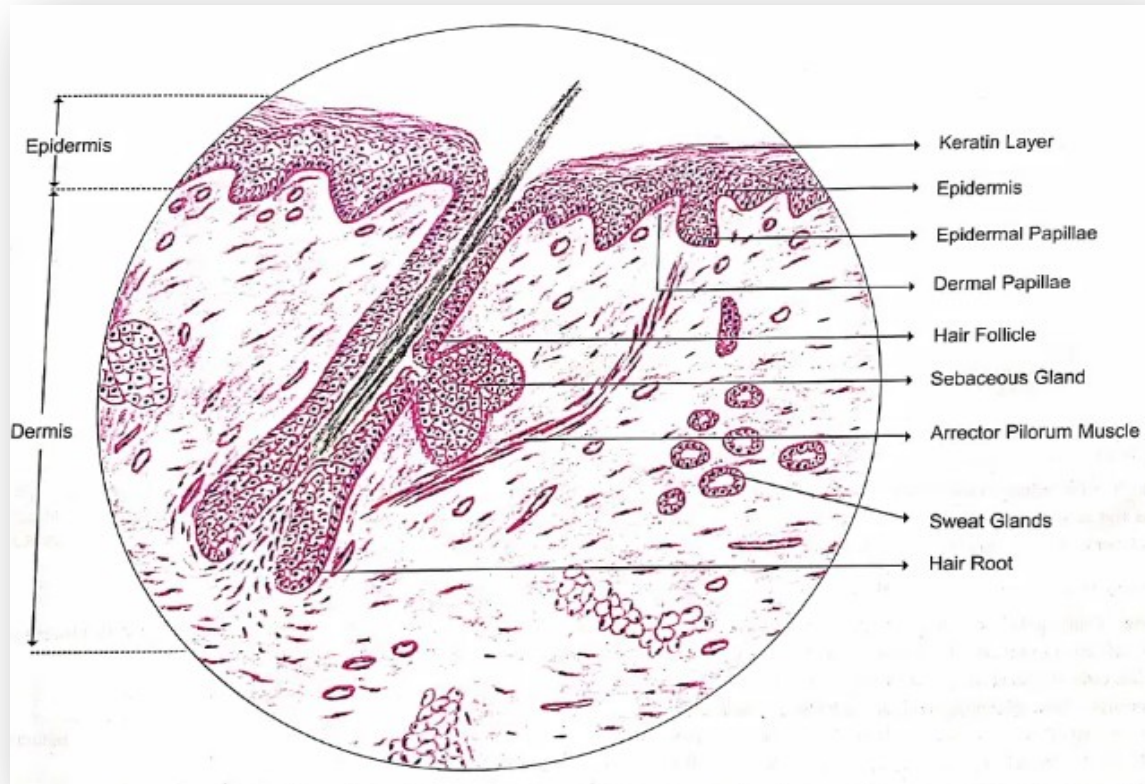
Lamina propria - intervening connective tissue between the anterior and posterior surfaces

Tarsal plate - a sheet of dense connective tissue (fibro-elastic) in the posterior region of the lamina propria.

Meibomian glands - sebaceous glands found within the tarsal plate.

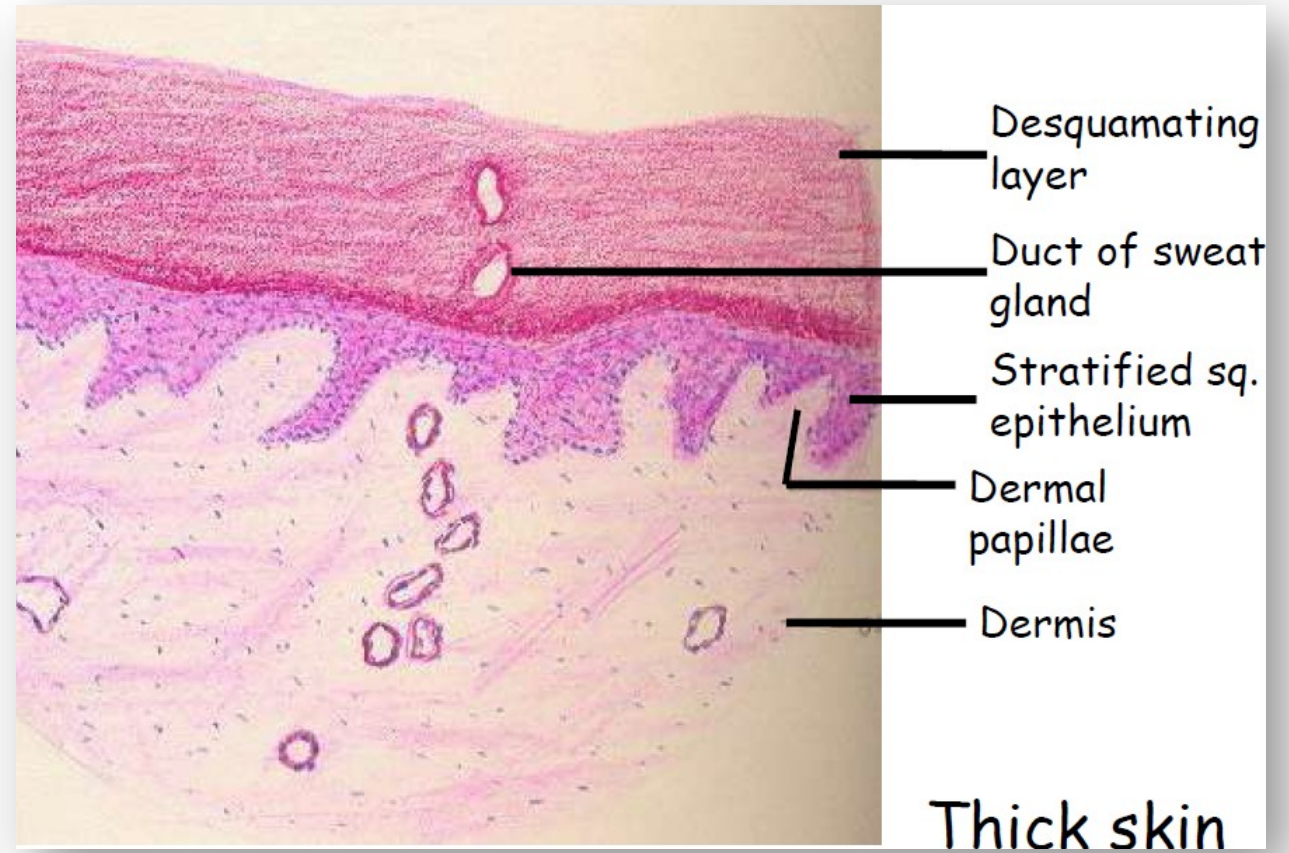
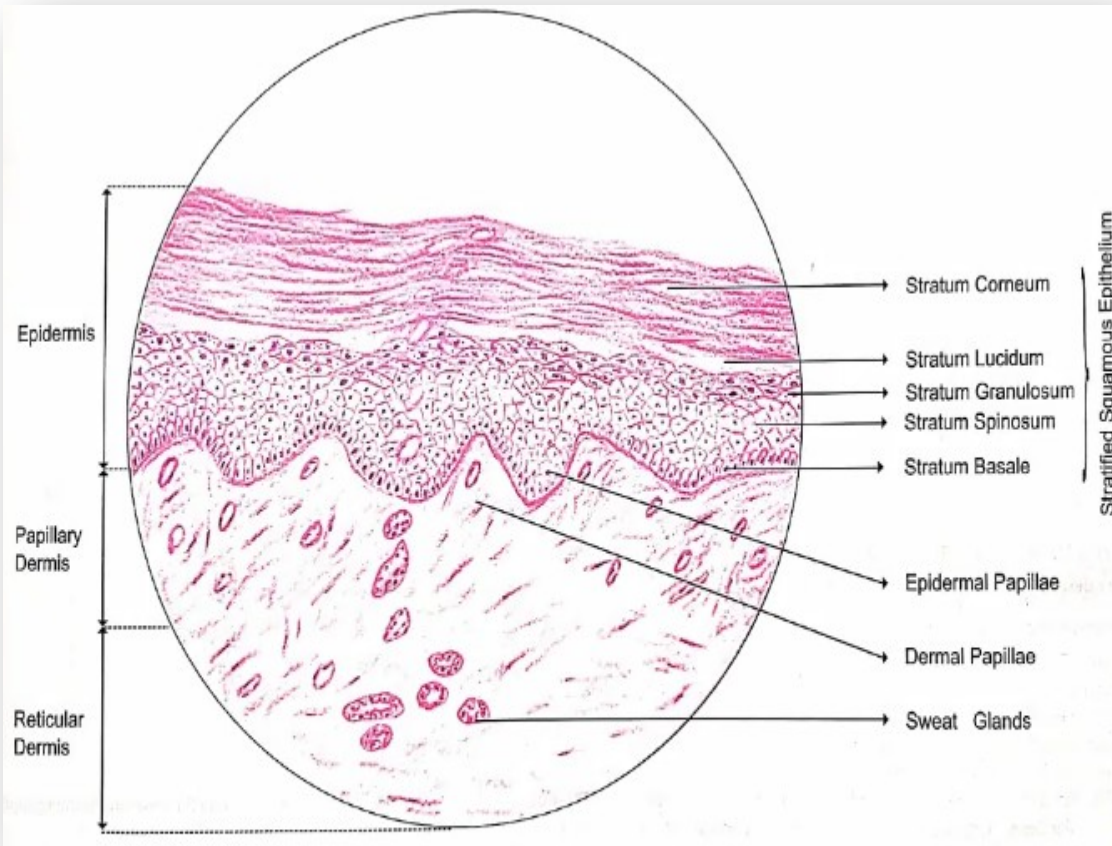


Thin skin



Thin skin

Thick skin



Eye lid

